Government of Telangana
Hyderabad Metropolitan Water Supply and Sewerage Board
(HMWSSB)

Telangana State

OSMAN SAGAR DAM
Comprehensive Dam Safety Evaluation Report

November/ 2025

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Chapter 1: Regulatory and Legal Framework

1.1 Provision under the Dam Safety Act-2021

1.1.1 Section 38

(1) The owner of a specified dam shall make or cause to be made comprehensive dam safety evaluation of each specified dam through an **independent panel of experts** constituted as per regulations for the purpose of **determining the conditions of the specified dam and its reservoir**:

Provided that the **first comprehensive dam safety evaluation for each existing specified dam shall be conducted within five years from the date of commencement of this Act**, and thereafter the comprehensive dam safety evaluation of each such dam shall be carried out at regular intervals as may be specified by the regulations.

(2) – The comprehensive dam safety evaluation shall consists of, but not be limited to,—

(a) Review and analysis of available data on the **design, construction, operation, maintenance and performance** of the structure.

(b) General assessment of **hydrologic and hydraulic conditions** with mandatory review of **design floods** as specified by the regulations.

(c) General assessment of **seismic safety** of specified dam with mandatory **site-specific seismic parameters study** in certain cases as specified by the regulations;

(d) Evaluation of the **operation, maintenance and inspection procedures**; and

(e) Evaluation of **any other conditions which constitute a hazard** to the integrity of the structure

1.1.2 Section 39

The comprehensive dam safety evaluation referred to in section 38 shall be compulsory in the case of,—

(a) **Major modification** to the original structure or design criteria.

(b) **Discovery of an unusual condition** at the dam or reservoir rim; and

(c) **An extreme hydrological or seismic** event.

1.1.3 Section 40

(1) The owner of a specified dam **shall report the results of the dam safety evaluation** undertaken under section 38 or section 39 to the State Dam Safety Organization.

(2) – The reports referred to in sub-section (1) shall include, but not be limited to,—

(a) **Assessment of the condition of the structure** based on the visual observations and available data on the **design, hydrology, construction, operation, maintenance** and performance of the structure.

(b) **Recommendations for any emergency measures** or actions, if required, to assure the immediate safety of the structure.

(c) **Recommendations for remedial measures and actions** related to design, construction, operation, maintenance and inspection of the structure, if required.

(d) Recommendations for additional detailed studies, investigations and analysis, if required; and

(e) Recommendations for improvements in routine maintenance and inspection of dam, if required.

(3) – Where the safety evaluations undertaken under section 38 or section 39, results in recommendations for a remedial action, the **State Dam Safety Organization shall pursue with the owner** of the specified dam **to ensure that remedial measures** are carried out in time, for which the owner shall provide adequate funds.

(4) – Where there is any **unresolved matter emerging between an independent panel of experts** referred to in sub section (1) of section 38 **and the owner** of the specified dam, the matter shall be **referred to the State Dam Safety Organization**, and, in case no agreement is arrived at, the matter shall be referred to the Authority which shall render its advice and send recommendations to the State Government concerned for implementation.

1.2 Regulations related to Comprehensive Dam safety Evaluation made under the Act

Enclosed in Annexure

Dam Safety Act 2021, Section 54 (p) - Extraordinary Gazette No. 314,
(8) Time interval for CDSE of Dams.

Dam Safety Act 2021, Section 54 (r) - Extraordinary Gazette No. 314,
(9) Mandatory Site Specific Seismic Parameters studies of Specified Dams.

Dam Safety Act 2021, Section 54 (q) - Extraordinary Gazette No. 173,
(8) Review of Design Flood of existing Dam.

Chapter 2: Constitution of Independent Panel of Expert

Regulations to constitute the independent panel of expert areenclosed as Annexure.

Chapter 3: Checklist of components of the Project Evaluated

Component Inspected	Tick the box
EMBANKMENT DAM	
Upstream Slope	
Downstream Slope	
Abutments	
Crest	
Seepage Areas	
Internal Drainage	
Relief Drains	
CONCRETE DAM	
Upstream Face	
Downstream Face	
Abutments	
Crests	
SPILLWAYS	
Approach Channel	
Stilling Basin	
Discharge Channel	
Control Features	
Erosion Protection	
Side Slopes	
INLETS, OUTLETS AND DRAINS	
Inlet & Outlets	
Stilling Basin	
Discharge Channel	
Trashracks	
Emergency Systems	
GENERAL AREAS	
Reservoir Surface	
Shoreline	
Mechanical Systems	
Electrical Systems	
Upstream Watershed	
Downstream Flood-Plains	

Chapter 4: Project details, salient features of the dam and its background history

4.1 Projectdetails

1. Project Description:

a.Project Identification Code (PIC): TL47HH0018

(As given in National Register of Specified Dams)

b. Project Name: Osmansagar

c. River Basin Krishna

d. Sub River Basin: Musi ı: Musi

2. Project Location:

a.State: Telangana b. District: Rangareddy

c. Earthquake Zone: Zone-II

d. Survey of India Topo Sheet No. E44M7

e. Nearest City: Hyderabad f. Nearest Airport: RGIA Shamshabad

g. Nearest Railhead:Nampally

h. Name of Immediate u/s Project: None

i. Name of Immediate d/s Project: Musi

j. Latitude/Longitude (in degrees, minutes, seconds):

Lat:	17	2	43.6	N
Long:	78	18	59.3	E

3. Project Benefits:

a. Type of Project: Flood control and Drinking Water

b. Irrigation Benefits, in hectares (ha):

(i) Gross Command Area (GCA): 0

(ii) Cultivable Command Area (CCA): 0

(iii) Annual Irrigation Potential (AIP): 0

c. Hydropower Benefits:

(i) Installed Capacity (MW): 0 (ii) Firm Power (MW): 0

(iii) Average Annual Energy Generation (MU): 0

d. Domestic/Municipal/Industrial Water Supply:

(i) Annual Water Supply (MCM): 45

(ii) Nos. of Population Benefitted (In Lakh): 9.5

e. Flood Protection:

(i) Flood Protected Area (ha): 30000

(ii) Population Benefitted (No): 30,00,000

f. Details of Tourism/Recreational Facilities: Park

4. Project Ownership Details:

a. Dam Owning Agency: HMWSSB

b. Implementing Agency: Transmission Division-I

c. Details of Dam Incharge:

(i) Name: Mr.D.Sudarshan

(ii) Designation: Director-Technical

(iii) Phone No. (With STD Code): 040 23442844 (Ext - 500)

(iv) Fax No. --

(v) E-mail: dtranshmwssb@hyderabadwater.gov.in

(vi) Contact Address: 5th Floor, HMWSSB Admin Building, Khairathabad, Hyderabad-500003

4.2 Salient features of the dam

Dam Features:
I. **Main Dam**
a. Type: Masonry+Composite
b. Total length of the Main dam (m): 1920
c. Length of Composite dam (m): 884
d. Length of Masonry dam (m): 1036
e. Top width of Composite Dam (m): 10.668
f. Top width of Masonry Dam (m):6.096
g. Elevation of top of Composite Dam (m):……
h. Elevation of top of Masonry Dam (m): ……
i. Elevation of top of Upstream Solid Parapet Wall (m): ……
j. Height of Composite Dam above Lowest River Bed Level (m): ……
k. Height of Masonry Dam above deepest foundation level (m): ……
l. Lowest River Bed Elevation (m): 535.53
m. Deepest Foundation Elevation (m): 35.97
II. **Saddle Dam(Natural Bye Wash)**
a. Type:
b. Length of the Saddle dam (m): 3 nos of length 332 m each
c. Top width of Saddle Dam (m):
d. Elevation of top of Saddle Dam (m): 550.164 m each
e. Elevation of top of Upstream Solid Parapet Wall (m):
f. Height of Saddle Dam above Lowest Bed Level
in case of embankment dam or above deepest
foundation level in case of concrete / masonry dam (m):

III **Main Spillway**:
(a) Type of Spillway: 1.Sluice type gated spillway @ + R.L 541.72 2.Open Vents @ R.L 546.65 3. Open Vents @ R.L 545.59
(b) Length of Spillway (m):1) 60 Mts 2) 60 Mts 3) 60 Mts
(d) Location of Spillway: Central Spill (Ch No 44 to 46) – 2nos Right Flank (Ch No 58 to 60) – 1 no
(Central Spillway/Left Flank/Right Flank/Saddle, in addition Chainage may also be mentioned)
(d) Spillway Crest Level (m):1. Sluice type gated spillway @ + R.L 541.72 2.Open Vents @ R.L 546.65 3. Open Vents @ R.L 545.59
(e) Number of Bays/spillways: 1. Sluice type gated spillway – 15 nos 2.Open Vents – 15 nos 3. Open Vents - 15 nos
(f) Number and thickness of Piers (m): 1. Sluice type gated spillway – 14 Nos 2. Open Vents - 14 Nos 3. Open Vents – 14 Nos
(g) Total Discharging Capacity at FRL and MWL (m3/s): 429 and 2608
(h) Design head used for working out spillway crest profile (m): This is sluice type spillway
(i) Type of Energy Dissipation Arrangement: Discharge is getting released on bed rock
(j) Type of Spillway Gate: Slide cum guide roller vertical lift Gates
(k) Size of Spillway Gate: 1. Sluice type gated spillway (15 nos) - 1.83 W x 3.05 H (at R.L 541.72) 2.Open Vents (15 nos) – 1.22 W x 2.74 H (at R.L 546.65) 3. Open Vents (15 nos)- 1.83 W x 4.11 H(at R.L 545.59)
(l) Type of Hoist for Spillway Gates: Rope Drum
(Rope Drum/ Hydraulic)
(m) Hoist Capacity of Spillway Gates (MT): Not Available
(n): Hoist Operation:Through a manual
(Manual / Electrical / Remote Control)
(o) Number of Sets of Stop-logs: 1 no of bulk head gate is available
(p) Number of Stop Log Units per Set & Size: NA
(q) Number of Gantry Crane(s) for Stop Log Gates: 1

(r) Gantry Crane Capacity (MT): NA
IV Auxiliary Spillway:
(a) Type of Spillway 1.Natural Bye Wash– 3 nos(R.L 550.20). Currently Non functional
(b) Length of Spillway (m):Natural Bye Wash– 3 nos X 332.3
(c) Location of Spillway: Natural Bye Wash– 3 nos – not traceable (R.L 550.20)
(Central Spillway/Left Flank/Right Flank/Saddle, in addition Chainage may also be mentioned)
(d) Spillway Crest Level (m) Natural Bye Wash (3 nos) - R.L 550.20
(e) Number of Bays/Spillways: Natural Bye Wash (3 nos)
(f) Number and Thickness of Piers: Not traceble
(f) Total Discharging Capacity at FRL &MWL (m3/s): At FRL -Nil MWL – 2608(as per Original Design)
(h) Design head used for working out spillway crest profile (m): Not Available
(i) Type of Energy Dissipation Arrangement: Natural bed rock
(j) Type of Spillway Gate: Ungated
(k) Size of Spillway Gate: Width (m) NA Height (m) NA
(l) Type of Hoist for Spillway Gates: NA
(m) Hoist Capacity of Spillway Gates (MT): NA
(n): Hoist Operation: NA
(Manual/Electrical/Remote Control)
(o) Number of sets of Stop-logs: NA
(p) Number of Stop Log Units per set & size: NA
(q) Number of Gantry Crane(s) for Stop Log Gates: NA
(r) Gantry Crane Capacity (MT): NA
V. Fuse Plug: No Fuse Plug provided
(a) Location: NA

(b) Length (m): NA
(c) Crest Level (m): NA
(d) Top Width (m): NA
(e) Discharging Capacity at MWL (m3/s): NA
VI. Sluice Arrangement (In Concrete and Masonry Dams):
(a) No. of Sluices & Sill Level (m):
(b) Size of Sluice: Width (m): 1.524 Height (m): 1.372 Dia. (m): -
(c) Discharging Capacity of Sluice at FRL (m3/s): 6.27
(d) Type of Service Gate: Slide type Vertical lift gate
(e) Size of Service Gate: Width (m) Height (m)
(f) Type of Hoist for Service Gates:Screw rod hoist
(g) Hoist Capacity of Service Gates (M.T.):
(h): Hoist Operation:Manual
(Manual/Electrical/Remote Control)
(i) Type of Emergency Gate: Slide type Vertical lift gate
(j) Size of Emergency Gate: Width (m): 1.524 Height (m): 1.372 Dia. (m): -
(k) Type of Hoist for Emergency Gates: Screw rod hoist
(l) Hoist Capacity of Emergency Gates (M.T): Not available
(m): Hoist Operation: Manual
(Manual / Electrical)
VII. Outlet works (In Embankment, Concrete & Masonry Dams): Not Available
(a) Location: Chainage no- 29
(b) Number: 1 no
(c) Sill level (m): 535.53
(d) Size: Width (m): 1.524 Height (m): 1.372 Dia. (m): -
(e) Discharging Capacity (m3/s):6.27 (at FRL)

(f) Type of Service Gate: Slide type Vertical lift gate
(g) Size of Service Gate: Width (m): 1.524 Height (m): 1.372 Dia. (m): -
(h) Type of Hoist for Service Gates : Screw rod hoist
(i) Hoist Capacity of Service Gates (M.T): Not Available
(j) Hoist Operation:(Manual/Electrical/Both) Manual
(k) Type of Emergency Gate: Slide type Vertical lift gate
(l) Size of Emergency Gate: Width (m): 1.524 Height (m): 1.372 Dia. (m): -
(m) Type of Hoist for Emergency Gates : Slide type Vertical lift gate
(n) Hoist Capacity of Emergency Gates (M.T): Not Available
(o) Hoist Operation: Manual
(Manual / Electrical)
Reservoir Features:
a. Catchment Area at Dam site (km2): 738.15 b. Maximum Water Level (m): 551.69
c. Full Reservoir Level (m): 545.592
d. Minimum Draw Down Level (m): 535.55 e. Dead Storage Level (m): 534.01
f. Live Storage Capacity (Mm3): 97.923 (as per Hydrographic Survey 2024)
g. Gross Storage Capacity (Mm3) at FRL: 99.605 (as per Hydrographic Survey 2024)
h. Reservoir Spread Area (km2) at FRL: 24.79
3. Construction Aspects:
a. Date of Starting the Construction (DD/MM/YYYY): 1912
b. Date of Completion (DD/MM/YYYY): 1920
c. Designing Agency: Designs were made under the guidance of Sir Mokshagundam Visvesvaraya
d. Construction Agency: His Exalted Highness Nizam's Public Works Department
e. Construction Cost (Rupees in Lakh): Rs.60 Lakhs
4. Operational Aspects:
a. Date of first full impoundment (MM/YYYY): The project was commissioned in 1920
b. Whether Pre & Post monsoon inspection being carried out: Yes

c. Major recommendations of dam safety inspection, along with brief status on compliance: copy of last PoE done on 14.11.2022.
Major Recommendations:

1. Downstream of the bund is covered with vegetation of big trees. Due to this the condition of the downstream slope of high dams such as drains, rock toe, toe drain could not be inspected.

2. The surplus arrangements on the downstream flood plain may be made for free passage of designed discharge.

3. Substantial water is observed downstream of masonry dam. The source of leakage could not be assessed. It is suggested to observe the flow pattern at different reservoir levels by incorporating ‘ V ’ notches.

Hydromechanical:

1. All the structural components of gates are to be checked deficiency to be rectified.
2. Proper illumination to be provided.
3. Gantry crane needs repairs.
4. Operational and maintenance records are to be maintained.

d. Any operational failure in the past: No operational failures recorded.

e. Any other past dam incident: No past dam incident recorded.

f. Operation and Maintenance Manual: Yes (old)but new O&M manual is under preparation.

Year of publication:

g. Emergency Action Plan: No

Year of Publication:

5. Instrumentation Aspects:

(Data Records and other information including pictures can be included in Appendix II-D)

a. List of Instruments installed in the Dam:

Sl. No.	Name of Instrument	Working Status	Year of Installation	Nos of Year data available
1.	Water Level Sensor	Y/N	Y	
2.	Plumb Bob	Y/N	N	
3.	Inclinometer	Y/N	N	
4.	Stress meters	Y/N	N	

5.	Strain meters	Y/N	N
6.	Toe Drain	Y/N	N
7.	Drain Wells	Y/N	N
8.	V-Notches	Y/N	N
9.	Pressure Gauges	Y/N	N
10.	Accelerograph	Y/N	N
11.	SCADA	Y/N	N
12.	Surveillance	Y/N	N
13.	Rain Gauge ORG	Y/N	Y
14.	Rain Gauge SRRG	Y/N	Y
15.	………………		

d. Summary on adequacy and justification for additional instrumentation:

1.Toe drain to be restored at the discharging locations.

2. V-notches should be provided at discharge points.

3. Piezo meters and accelerograph are required to be installed.

4.3 Background history of the dam

Add comprehensive history of modifications, incidents, and major events affecting the dam.

Include review of the status of previous evaluation recommendations and completion status.

1. No major incidents/events recorded. No structural changes made.
2. Gantry crane repaired and made functional.
3. All the exposed structural members painted.
4. Illumination provided at necessary locations.
5. Hydrographic survey was conducted in the year 2024.
6. Permission for felling of trees obtained from forest department. Cutting of trees started, work is in progress.

Chapter 5: Inspection and review of different components

5.1 Ungated Spillway -

5.1.1 Type of ungated spillway

1. Open Vents – 15 nos (R.L 546.65)
2. Open Vents -15 nos (R.L 545.59)
3. Natural Bye Wash– 3 nos (R.L 550.20)

5.1.2 Review of design data

Sl No	Particular	Design Stage value	Current Value	Remarks
1	Design Inflow Value (Cumec) (PMF/SPF/Return Period Flood)	1886(Return Period of Floodtaken originally is not known).	6137(This is total design discharge as per flood review which includes gated spillways also. Separate values for ungated and gated not available. PMF condition is considered in the review of design flood.)	
2	Design Outflow value (Cumec) considering Routing effect	1886	1886	
3	Overflow spillway length (m)	378	378	
4	Free board (m)	1.5	1.5	
5	Max observed outflow (cumec)	____	6137	
6	Design Seismic parameter	Not Available	As per Zone-II	
7	Current Seismic parameter as per latest IS code/ NDSA Regulations	Not Available	To be evaluated	
8	Maximum Tail Water Level	Not Available	Not Available	

5.1.3 Spillway body material:

Stone Masonry

5.1.3.1 Observations on the body material - Observed and in good condition

5.1.3.2 Repercussions, if the issue is not addressed, if any - NA

5.1.3.3 Recommendations to address the issue, if any - Nil

5.1.4 Spillway Location:

Main Spillway:
Flood Gates - 15 nos(R.L 540.74) from Ch.no – 44 to 46

Auxiliary Spillway:

1. Open Vents – 15 nos (R.L 546.65) from Ch.no – 44 to 46
2. Open Vents -15 nos (R.L 545.59) from Ch.no – 58 to 60
3. Natural Bye Wash– 3 nos (R.L 550.20) location is not traceable.

5.1.5 Spillway crest:

5.1.5.1 Observations on the crest – observed and found in good condition

5.1.5.2 Repercussions, if the issue is not addressed, if any. - NA

5.1.5.3 Recommendations on the crest, if any - NA

5.1.6 Spillway downstream profile:

5.1.6.1 Observations on the downstream profile – a scour has developed immediately downstream of the training wall.

5.1.6.2 Repercussions, if the issue is not addressed, if any – if not attended this scour area will become deeper and further damage to the downstream flow will occur

5.1.6.3 Recommendations on the downstream profile, if any. - This scoured area to be filled with high performance concrete (M30). This concrete to be provided with anchors 20mm dia, 1 mtr long both towards bed rock and the concrete@ 1.5m C/c in three rows in staggered manner.

5.1.7 Energy Dissipation Arrangement:

5.1.7.1 Observations on the Energy Dissipation Arrangement

Generally, in good condition except the scour pocket as numerated in Para 5.1.6.1.

5.1.7.2 Repercussions, if the issue is not addressed, if any. - if not attended this scour area will become deeper and further damage to the downstream flow will occur

5.1.7.3 Recommendations on the Energy Dissipation Arrangement, if any - This scoured area to be filled with high performance concrete (M30). This concrete to be provided with anchors 20mm dia, 1 mtr long both towards bed rock and the concrete @ 1.5m C/c in three rows in staggered manner.

5.2 Gated Spillway

5.2.1 Type of gated spillway

Such as Ogee/Chute etc.

5.2.2 Review of design data

Sl No	Particular	Design Stage value	Current Value	Remarks
1	Design Inflow Value (Cumec) (PMF/SPF/Return Period Flood)	722.08	6137 (This is total design discharge as per flood review which includes gated spill ways also. Separate values for ungated and gated not available. PMF condition is considered in the review of design flood.)	
2	Design Outflow value (Cumec) considering Routing effect	2608.00	6137 (This is total design discharge as per flood review which includes gated spill ways	

			also. Separate values for ungated and gated not available. PMF condition is considered in the review of design flood.)	
3	Functional gates (No)	15	15	
4	Free board (m)	1.50	1.50	
5	Max observed outflow (cumec)	378.250	375.280	
6	Design Seismic parameter	Not Known	As per Zone-II	
7	Current Seismic parameter as per latest IS code/ NDSA Regulations	Not Known	Yet to be evaluated.	
8	Availability of stoplog gates	01	01	
9	Functional River Sluice Gates	15	15	
10	Maximum Tail Water level	Not measured	Not measured.	
11	Full Reservoir Level (m)	545.50	545.50	

5.2.3 Spillway body material

Masonry

5.2.3.1 Observations on the body material – in Good condition

5.2.3.2 Repercussions, if the issue is not addressed, if any.- NA

5.2.3.3 Recommendations on the body material, if any. - NA

5.2.4 Spillway Location:

5.2.4.1 Observations on the Spillway Location -

Within Main dam from Ch. 44 to Ch.46.

5.2.5 Spillway crest/Bridge deck:

5.2.5.1 Observations on the Spillway crest

Good Condition.

5.2.5.2 Repercussions, if the issue is not addressed, if any.- NA

5.2.5.3 Recommendations on the Spillway crest, if any. - NA

5.2.6 Spillway downstream profile:

5.2.6.1 Observations on the Spillway downstream profile– a scour has developed immediately downstream of the training wall.

5.2.6.2 Repercussions, if the issue is not addressed, if any.– if not attended this scour area will become deeper and further damage to the downstream flow will occur

5.2.6.3 Recommendations on the downstream profile, if any. - This scoured area to be filled with high performance concrete (M30). This concrete to be provided with anchors 20mm dia, 1 mtr long both towards bed rock and the concrete @ 1.5m C/c in three rows in staggered manner.

5.2.7 Energy Dissipation Arrangement:

5.2.7.1 Observations on the Energy Dissipation Arrangement-

Generally, in good condition except the scour pocket as numerated in Para 5.2.6.1.

5.2.7.2 Repercussions, if the issue is not addressed, if any.- if not attended this scour area will become deeper and further damage to the downstream flow will occur.

5.2.7.3 Recommendations on the Energy Dissipation Arrangement, if any.- This scoured area to be filled with high performance concrete (M30). This concrete to be provided with anchors 20mm dia, 1 mtr long both towards bed rock and the concrete @ 1.5m C/c in three rows in staggered manner.

5.2.8 Gate Hoisting Arrangement:

5.2.8.1 Observations on the Gate Hoisting Arrangement

1. The rope drum hoist mechanism is with one pulley provided over the gate at centre, the rope fixed on either side of the drum. The rope winding is on either side of single rope drum. The spur gear mechanism is provided with hand operation.

2. Over the years, during floods and more number of operations, the lifting arrangements with hoist mechanism, hoist supporting structure might have been affected.

3. The ropes are to be checked for any sagging and broken strands.

5.2.8.2 Repercussions, if the issue is not addressed, if any.– if the deficiencies are not attended in time, there will be operation troubles in gates.

5.2.8.3 Recommendations on the Gate Hoisting Arrangement, if any.

1. The pulleys positions with respect to cheek plates need to be checked along with shafts, bush bearings etc.Any deviations, if found, are to be attended. Lubrication is be carried out periodically.
2. The rope drum,drum shafts, supporting plates, tightness and sound ness of bolts and rope clamps to be checked, deviations to be attended.
3. Drum gear bolts to be checked for tightness and soundness. the misalignments of gears to be checked and adjusted. Ensure uniform meshing of teeth.
4. On the hoist supporting structure, rusting and pitting to be cleaned. The structural members are to be checked.The weak members, if found, are to be strengthened or replaced. Foundation bolts to be checked.The damaged bolts, if found, to be replaced.
5. The hoist bridge members, gantry rails, chequered plates, railings are to be checked for corrosion, rusting and pitting periodically. The weak sections, if found, are to be repaired/replaced. The water stagnation over the hoist bridge to be avoided.
6. The braking arrangements are to be checked for intermediate stops.
7. The weak ropes are to be replaced.

5.2.9 Gantry crane Arrangement:

5.2.9.1 Observations on the Gantry crane Arrangement

The gantry crane available at site is manually operated. The staff reported that the gantry crane repairs are taken up and trial runs will be made shortly.

5.2.9.2 Repercussions, if the issue is not addressed, if any.

The gantry crane is to be repaired and put into use then only repairs to main gates, bulk head can be attended.

5.2.9.3 Recommendations on the Gantry crane Arrangement, if any.

It is reported that the bulk head gate have been repaired. However, it could not be checked by lowering due to the reason that the crane was not operational because it is under repair. The bulk head gate to be lowered to attend main gate repairs replacements and maintenance.

5.2.10 Lifting beam & Counter weight

5.2.10.1 Observations on the Lifting beam & Counter weight

The lifting beam is not use since a long period.

5.2.10.2 Repercussions, if the issue is not addressed, if any.– If the lifting beam is having any operational problem, the bulk head operations will get affected.

5.2.10.3 Recommendations on the Lifting beam & Counter weight, if any.

Repairs if required are to be attended.

5.2.11 Trunnion beam & Walkway – There is no Trunnion beam & Walkway.

5.2.11.1 Observations on the Trunnion beam & Walkway

5.2.11.2 Repercussions, if the issue is not addressed, if any.

5.2.11.3 Recommendations on the Trunnion beam & Walkway, if any.

5.2.12 Gates, seals and embedded parts

5.2.12.1 Observations on the Gates, seals and embedded parts

1. This year the gates are operated more number of times during floods compared to past few years. The field staff reported that the maximum flood gate opening in the last 60 years occurred on 26.09.2025 from 9.00 AM to 8.00 PM and on the following day 15gates were opened to a height of 9 Feet. The moving components might have got affected due to these operations and heavy flood discharges.
2. These gates over 100 years old. The gates are with corrosion, rusting and pitting sparsely.
3. The skin plate, skin plate joints, joints between skin plates and horizontal girders, vertical stiffeners in wet condition if exposed to atmosphere are subjected to corrosion and rusting. Theforeign materials, particularly the sharp edge debris, uneven surface of seal seat,uneven loads may cause damages. The bottom seal damages due to uneven contact of seal and seal seat, loose fixing bolts. The seals might be affected.
4. The roller train supporting plates, lock plates, rollers, pins, bushes will get affected due to uneven forces. The rollers assemblies might have been affected. The rollers without maintenance will get jammed.
5. The horizontal girder with water stagnation, in wet condition exposed to atmosphere causes rusting and pitting.
6. The metal housing embedded, the roller path, seal seat, pad seat might have affected due to accumulation of foreign material, debris, trash.

7. The metal housing (groove) facilitate roller movement, seal seat,pad seat might have been affected heavy floods, more operation and being old.

5.2.12.2 Repercussions, if the issue is not addressed, if any. – if the deficiencies

5.2.12.3 are not timely attended, will affect the smooth operation of gates.

5.2.12.4 Recommendations on the Gates, seals and embedded parts, if any.

The Gates to be raised and brought to the inspection chambers one by one and to be attended as given below

1. The skin plate and skin plate joints to be checked for corrosion, rusting and pitting. In case there are pittings on the skin plate, they are to be filled with weld and grinding is to be carried out to have smooth surface.
2. The gates and its structural components are not painted for more than 5 years because of non-availability of bulk head gate. The Gates need to be raised and brought to the inspection chamber one by one and got painted with as per BIS14177.
3. The bottom seals, side seals and top seals are to be checked, the damaged seals and seal fixing components are to be replaced. Ensure proper contact as per standards.
4. The bottom seals, seal packing plates, bolts are to be checked, deviations if found are to be rectified, the damaged items to be rectified. Till repairs are carried out as detailed above, the present leakages are to be arrested by caulking.
5. The joints of skin plates, joints between skin plate and Horizontal girder, verticals to be checked and rectified.
6. The rollers are to be cleaned and checked for smooth functioning and they are to be checked with hand for free rotation. The jammed rollers are to be checked, deviation in rollers if found to be attended, the damaged rollers and bearings are to be replaced. The periodical lubrication of rollers is very much required as they bear and transmit the load through the roller track to the civil structure.
7. The horizontal girders supporting the skin plate are the main load carrying members, they are to be checked for corrosion, rusting, pitting, weak portions. The joints between horizontal girders and skin plates and verticals are to be checked. Deviations if found to be attended.
8. The water stagnations on webs to be attended regularly. The vertical members are also to be checked and deviations to be attended. Care shall be taken to avoid failure due to weak sections.
9. The end girder receive the water thrust from horizontal girders and transmit to rollers. The end girder assembly bear and transmit the load to pad and pad seat. The damages, verticality, weak joints to be checked. The weak portions if found, to be strengthened, deviation in verticality to be attended .Ensure that the rocker arm facilitates perfect seating and transmission of load to rollers. The rocker arms are to be lubricated properly and regularly to have smooth operation.
10. The groove made of metal housings are to be checked for any damages, verticality and uniform surfaces, in case of any deviations, they are to be attended. The seal tracks

roller tracks are to be checked and ensure proper seating of seals and rollers over their tracks. The obstructions if any in the grooves which prevents a smooth gate movement are to be removed and cleaned.

11. The sill beams are to be checked for horizontality and proper seating of bottom seal, obstructions due to debris and silt deposits, the required rectifications if needed are to be attended, obstructions to be removed and maintained neatly.
12. During the dry sill condition(or) by lowering bulk head gate, the gates are to be inspected in detail, the minor field repairs and replacements if required to be taken up along with periodical maintenance, in case of major repairs and replacements, they are to be taken up and completed before floods. These are the age old gates and require regular maintenance, repairs and replacements to keep them in operable condition.

5.2.13 River Sluice Gates, seals and embedded parts

There are no River sluices in the project.

5.2.13.1 Observations on the River Sluice Gates, seals and embedded parts, if any.

5.2.13.2 Repercussions, if the issue is not addressed, if any.

5.2.13.3 Recommendations on the River Sluice Gates, seals and embedded parts, if any.

5.2.14 Outlet Gate

The sluice is provided with one vent of size 5’X4.5’, where in emergency and service gates are existing (vertical lift gates). The gates are operated with screw rod hoist mechanism. The service gate hoist mechanism is the worm and worm wheel and screw rod. The emergency gate is provided with simple screw rod and nut hoist mechanism.

5.2.14.1 Observations on the Outlet Gate, if any.

Both the emergency and service gates are under water. So their condition couldn’t be assessed.

5.2.14.2 Repercussions, if the issue is not addressed, if any.

In the absence of regular maintenance, the components will get affected due to rusting and pitting and may lose design strength ultimately will fail.

5.2.14.3 Recommendations on the Outlet gates, if any.

1. The condition of the gates will be assessed during slack season and the necessary checks and action may be taken as under.
2. The skin plates are to be checked for corrosion rusting and pitting, deviations if found to be attended. The seals are to be checked, the damaged seals and seal components are

to be replaced. The seals are to be checked for proper seating over seal seats. The verticality and uniform surface of seal tracks /seats to be checked and maintained.

3. The joints between skin plates and horizontal girders, vertical members, roller housings to be checked deviations if found to be attended.
4. Sliding metal pads and pad seats are to be checked for verticality uniform surface.
5. During periodical maintenance, moving components are to be lubricated to have smooth movement.
6. The grooves are to be checked for any obstructions (trash, debris) and damages. The pad seats seal tracks are to be checked for verticality and uniform surface. The obstructions are to be removed, damages are to be attended. The deviations if found are to be attended to have proper seating of seals and rollers over seal tracks and roller tracks.
7. In case non availability of dry sill condition, the emergency gate to be lowered and closed to attend maintenance and repairs and replacements of service gate.
8. The service gate hoist mechanism consists of screw rod, gear box with worm and worm wheel mechanism. The worm and worm wheel alignments are to be checked and adjusted to have proper meshing of worm threads and worm wheel teeth; they are to be lubricated to have smooth operation.
9. The hoist bridge members (main members, cross members, chequered plates, railings etc) to be checked, the metal surfaces with corrosion, rusting, pitting to be attended. The stagnation of water over hoist bridge to be avoided.

5.3 Foundation gallery and other GalleriesNo Gallery

5.3.1 Review of design data

Sl No	Particular	Design Stage value	Current Value	Remarks
1	Accessibility by Lift	(Pl state whether the galleries were accessible by lift originally or not)	(Pl state whether the galleries are now accessible by lift or not)	
2	Availability of access from Downstream			
3	Functional Pumps available to drain out water (No)			
4	Capacity of all pumps (HP)			
5	Max observed Seepage flow in the galleries (cumec)			

5.3.2 Galleries body material:

Concrete/Masonry/Composite (Masonry+ Concrete)

5.3.2.1 Observations on the Galleries body material parts

5.3.2.2 Repercussions, if the issue is not addressed, if any.

5.3.2.3 Recommendations on the Galleries body material, if any.

5.3.3 Gallery Location:

5.3.3.1 Observations on the Gallery Location

consider Elevation (m)

5.3.4 Foundation drains:

5.3.4.1 Observations on the Foundation drains

(Consider spacing, condition (flowing and chocked (numbers), condition of connection to gallery drain, last cleaning date etc)

5.3.4.2 Repercussions, if the issue is not addressed, if any.

5.3.4.3 Recommendations on the Foundation drains, if any.

5.3.5 Block joint drains:

5.3.5.1 Observations on the Block joint drains

(Consider spacing, condition (flowing and chocked (numbers),condition of connection to gallery drain, last cleaning date etc)

5.3.5.2 Repercussions, if the issue is not addressed, if any.

5.3.5.3 Recommendations on the Block joint drains, if any.

5.3.6 Form drains:

5.3.6.1 Observations on the Form drains

(Consider spacing, condition (flowing and chocked (numbers)), condition of connection to gallery drain, last cleaning date etc)

5.3.6.2 Repercussions, if the issue is not addressed, if any.

5.3.6.3 Recommendations on the Form drains, if any.

5.3.7 General condition of gallery:

5.3.7.1 Observations on the gallery

(Consider access to gallery, easy movement in the gallery, lighting arrangement, ventilation, damage, seepage and leaching on walls and roof)

5.3.7.2 Repercussions, if the issue is not addressed, if any.

5.3.7.3 Recommendations on the gallery, if any.

5.3.8 Sump:

5.3.8.1 Observations on the Sump

(Consider capacity, pumping arrangement, protection)

5.3.8.2 Repercussions, if the issue is not addressed, if any.

5.3.8.3 Recommendations on the Sump, if any.

5.4 Non-Over Flow Masonry Block

5.4.1 Review of design data

Sl No	Particular	Design Stage value	Current Value	Remarks
1	Design Inflow Value (Cumec)	---	---	---
2	Design Outflow value (Cumec) considering Routing effect	---	---	---
3	Full Reservoir Level (m)	545.	---	---
4	Free board (m)	---	---	---
5	Design Seismic parameter		NA	
6	Current Seismic parameter as per latest IS code/ NDSA Regulations	NA		
7	Maximum Tail Water level			
8	Maximum Water Level			

5.4.2 Dam body material:

Concrete/Masonry/Composite (Masonry+ Concrete)

5.4.2.1 Observations on the Dam body material

5.4.2.2 Repercussions, if the issue is not addressed, if any.

5.4.2.3 Recommendations on the Dam body material, if any.

5.4.3 Crest:

5.4.3.1 Observations on the Crest

(Consider material damage, vegetation, blockage, settlement, parapet wall, control room location, Tilt etc)

5.4.3.2 Repercussions, if the issue is not addressed, if any.

5.4.3.3 Recommendations on the Crest, if any.

5.4.4 Downstream profile:

5.4.4.1 Observations on the downstream profile

(Consider profile, damage, vegetation, blockage, body Seepage, access to toe area etc)

5.4.4.2 Repercussions, if the issue is not addressed, if any.

5.4.4.3 Recommendations on the downstream profile, if any.

5.4.5 Stability of Block as per latest codal provision

5.4.5.1 Observations on the Stability of Block

5.4.5.2 Repercussions, if the issue is not addressed, if any.

5.4.5.3 Recommendations on the Stability of Block, if any.

5.5 Non-Over Flow Composite Block

5.5.1 Review of design data

Sl No	Particular	Design Stage value	Current Value	Remarks
1	Design Inflow Value (Cumec)	---	---	---
2	Design Outflow value (Cumec) considering Routing effect	---	---	---
3	Full Reservoir Level (m)	545.59 m	545.59 m	---
4	Free board (m)	1.5 m	To be evaluated	---
5	Design Seismic parameter	Not known	To be evaluated	
6	Current Seismic parameter as per latest IS code/ NDSA regulations	Not known	Zone-II	
7	Maximum Tail Water level	---	---	---
8	Maximum Water Level	---	---	---

5.5.2 Dam body material:

Composite (Earthen and Stone Masonry) and Stone Masonry

5.5.2.1 Observations on the Dam body material

1. The stone masonry is in good condition.
2. The earthen dam portion is fully covered with vegetation and trees and therefore could not be examined thoroughly. As per the dam authorities, no leakages/seepages observed through the body of the dam.

5.5.2.2 Repercussions, if the issue is not addressed, if any.

If the vegetation cover on the earthen dam portion and the earthen section is not brought to the safe design condition it's stability may be compromised.

5.5.2.3 Recommendations on the Dam body material, if any.

The soil properties of the earthen dam to be evaluated to confirm whether it is safe as per the present seismic and codal provisions.

5.5.3 Crest:

5.5.3.1 Observations on the Crest

It is in Good Condition. No control room is provided.

5.5.3.2 Repercussions, if the issue is not addressed, if any.

No serious issues

5.5.3.3 Recommendations on the crest, if any.

Nil

5.5.4 Downstream slope:

5.5.4.1 Observations on the Downstream slope

A. Earthen portion:

1. The earth slope is in very bad condition covered with thick vegetation and trees.
2. The slope at many places having tree stubs needs to be removed along with its roots.
3. The slope is not maintained as per the original design.
4. The chute drains are either damaged or missing.
5. Dam rock toe and toe drain are fully covered with debris.
6. No instrumentation is provided.
7. As the face is covered fully with vegetation, its condition could not be evaluated for any seepage taking place. However, no flowing water was observed.

B. Masonry portion:

The masonry is general in good condition, however the seepage marks are observed at some location because of rainwater falling on the dam top road and flowing down through the joint of the road and parapet wall.

5.5.4.2 Repercussions, if the issue is not addressed, if any.

If not attended to the stability of the earthen dam will be endangered.

5.5.4.3 Recommendations on the Downstream slope, if any.

A. **Earthen Portion**:

1. The vegetation to be cut and uprooted including stubs of the trees.
2. The trees to be cut and uprooted after obtaining permission from Forest Department
3. After carrying out the review of the stability, the earthen slope to be brought to the design section.
4. Damaged chutes to be repaired/re constructed.

5. Rock toe including toe drain be cleared off debris and made functional.
6. V-Notches to be provided at suitable locations for the measurement of seepage water.
7. Piezo meters to be installed at suitable locations as per relevant BIS code.

B.**Masonry portion**

The open joint between the dam top road and the parapet walls on both sides to be filled with a mixture of bitumen and sand made water tightening. Suitable arrangement to dispose off rain water falling on the top of the road to be made.

5.5.5 Upstream Slope:

5.5.5.1 Observations on the Upstream slope

There is no upstream slope as the dam is composite and the upstream is made of rock masonry.

The reservoir was at 545.59 meters. Therefore, the upstream face of the rock masonry could be inspected above this level. The condition of the masonry, is generallyin good condition. However, vegetation is seen at some locations.

5.5.5.2 Repercussions, if the issue is not addressed, if any.

The observations made are not that serious. But, aesthetically not looking good. Since it is a heritage structure maintaining heritage is also important.

5.5.5.3 Recommendations on the Upstream slope, if any.

The vegetations to be cut and uprooted.

5.5.6 Downstream toe drain:

5.5.6.1 Observations on the toe drain

1. Dam rock toe and toe drain are fully covered with debris.
2. Due to above reason no observation of seepage water are being carried out.
3. Due to the same reason, there is no access to the toe area.

5.5.6.2 Repercussions, if the issue is not addressed, if any.

If the Toe drains are not operational, there would be build up of pore water pressure leading to instability of the slope.

5.5.6.3 Recommendations on the toe drain, if any.

The rock toe and toe drain to be cleared off all the debris and made operational.

5.5.7 Recommendation on Possibility of Failure

There is no piping taking place in the dam. There is no possibility of the failure.

5.5.8 Filters :

Filters are not provided.

5.5.8.1 Observations on the Filter

N/A

5.5.8.2 Repercussions, if the issue is not addressed, if any.

N/A

5.5.8.3 Recommendations on the Filter, if any.

N/A

5.6 Barrage

There is no Barrage in the project.

5.6.1 Review of design data

Sl No	Particular	Design Stage value	Current Value	Remarks
1	Design Inflow Value (Cumec) Return Period Flood)			
2.	Number of Bays			
3	Functional gates (No)			
4	Free board (m)			
5	Max observed outflow (cumec)			
6	Design Seismic parameter			
7	Current Seismic parameter as per latest IS code/ NDSA Regulations			
8	Availability of stop log gates			
9	Pond Level (m)			
10	HFL (m)			

11	Maximum Tail Water level (m)			
12	Crest Level (m)			
13	River Bed level (m)			

5.6.2 Pier:

Concrete/Masonry/Composite (Masonry+ Concrete)

5.6.2.1 Observations on the material

5.6.2.2 Repercussions, if the issue is not addressed, if any.

5.6.2.3 Recommendations on the body material, if any.

5.6.3 Bridge deck:

5.6.3.1 Observations on the Bridge deck

(Consider length, material damage, vegetation, settlement, parapet wall, control room location, Tilt etc)

5.6.3.2 Repercussions, if the issue is not addressed, if any.

5.6.3.3 Recommendations on the Bridge deck, if any.

5.6.4 Foundation:

Raft/ Gravity Floor

5.6.4.1 Observations on the Foundation condition

5.6.4.2 Repercussions, if the issue is not addressed, if any.

5.6.4.3 Recommendations on the Raft, if any.

5.6.5 Energy Dissipation Arrangement:

5.6.5.1 Observations on the Energy Dissipation Arrangement

(Consider length, damage, vegetation, blockage, training wall, endsill/bucket, apron, Channel encroachment, training wall etc)

5.6.5.2 Repercussions, if the issue is not addressed, if any.

5.6.5.3 Recommendations on the Energy Dissipation Arrangement, if any.

5.6.6 Gate Hoisting Arrangement:

5.6.6.1 Observations on the Gate Hoisting Arrangement

(consider electrical/mechanical efficacy, general condition)

5.6.6.2 Repercussions, if the issue is not addressed, if any.

5.6.6.3 Recommendations on the Gate Hoisting Arrangement, if any.

5.6.7 Gantry crane Arrangement:

5.6.7.1 Observations on the Gantry crane Arrangement

(Consider electrical/mechanical efficacy, general condition)

5.6.7.2 Repercussions, if the issue is not addressed, if any.

5.6.7.3 Recommendations on the Gantry crane Arrangement, if any.

5.6.8 Lifting beam & Counter weight

5.6.8.1 Observations on the Lifting beam & Counter weight

(Consider general condition and efficacy)

5.6.8.2 Repercussions, if the issue is not addressed, if any.

5.6.8.3 Recommendations on the Lifting beam & Counter weight, if any.

5.6.9 Trunnion beam & Walkway

5.6.9.1 Observations on the Trunnion beam & Walkway, if any.

(Consider general condition and efficacy)

5.6.9.2 Repercussions, if the issue is not addressed, if any.

5.6.9.3 Recommendations on the Trunnion beam & Walkway, if any.

5.6.10 Gates, seals and embedded parts

5.6.10.1 Observations on the Gates, seals and embedded parts

(Consider general condition and efficacy)

5.6.10.2 Repercussions, if the issue is not addressed, if any.

5.6.10.3 Recommendations on the Gates, seals and embedded parts, if any.

5.7 Saddle Dam (Gated/Ungated)

The saddle dams are in the form of 3 Nos of natural byewash which are ungated

5.7.1 Review of design data

Sl No	Particular	Design Stage value	Current Value	Remarks
1	Design Inflow Value (Cumec)	1053.39	0	3 Nos of byewash are blocked.
2	Design Outflow value (Cumec) considering Routing effect	1053.39	0	
3	Full Reservoir Level (m)	545.59	545.59	
4	Free board (m)	N/A	N/A	
5	Design Seismic parameter	N/A	N/A	
6	Current Seismic parameter as per latest IS code/ NDSA Regulations	N/A	N/A	
7	Maximum Tail Water level	N/A	N/A	
8	Maximum Water Level	N/A	N/A	

5.7.2 Dam body material:

There is no Dam body. It was natural surplussing arrangement.

5.7.2.1 Observations on the Dam body material

N/A

5.7.2.2 Repercussions, if the issue is not addressed, if any.

If no alternate arrangement is made, there is builds up of reservoir level and water level will encroach the free board area and put the dam structure into stress.

5.7.2.3 Recommendations on the Dam body material, if any.

N/A

5.7.3 Crest:

5.7.3.1 Observations on the Crest

The crest blocked by the vegetation and habitation.

5.7.3.2 Repercussions, if the issue is not addressed, if any.

If no alternate arrangement is made, there is builds up of reservoir level and water level will encroach the free board area and put the dam structure into stress.

5.7.3.3 Recommendations on the crest, if any.

Alternate arrangement to spill the water to be made.

5.7.4 Downstream slope:

5.7.4.1 Observations on the Downstream slope

Currently the downstream slope is not existing.

5.7.4.2 Repercussions, if the issue is not addressed, if any.

If no alternate arrangement is made, there is builds up of reservoir level and water level will encroach the free board area and put the dam structure into stress.

5.7.4.3 Recommendations on the Downstream slope, if any.

Alternate arrangement to spill the water to be made.

5.7.5 Upstream Slope:

5.7.5.1 Observations on the Upstream slope

The slope is not approachable as high level wall constructed there.

5.7.5.2 Repercussions, if the issue is not addressed, if any.

If no alternate arrangement is made, there is builds up of reservoir level and water level will encroach the free board area and put the dam structure into stress.

5.7.5.3 Recommendations on the Upstream slope, if any.

Alternate arrangement to spill the water to be made.

5.7.6 Downstream toe drain:

5.7.6.1 Observations on the toe drain

As the area is covered with vegetation and Habitation, toe area could not be approached.

5.7.6.2 Repercussions, if the issue is not addressed, if any.

N/A

5.7.6.3 Recommendations on the toe drain, if any.

N/A

5.7.7 Stability of Block as per latest codal provision.

5.7.7.1 Observations on the Stability of Block

The structural details including its profile is not available. Therefore no comment in this regard can be made.

5.7.7.2 Repercussions, if the issue is not addressed, if any.

5.7.7.3 Recommendations on the Stability of Block, if any.

5.7.8 Recommendation on Possibility of Failure

The structure is not available, therefore no observations in this regard can be made.

5.7.9 Filters

5.7.9.1 Observations on the Filter

No comment.

5.7.9.2 Repercussions, if the issue is not addressed, if any.

5.7.9.3 Recommendations on the Filter, if any.

5.8 Instrumentations of the Dam

The list of mandatory instruments as per the Regulations published vide Gazette Notification dated 24.04.2024 is enclosed as Annexure-

5.8.1 Summary of instruments installed with data availability and performance.

Sl. No.	Name of Instrument	Minimum number required as per the Regulations of DSA-2021	No. of instrument Installed at Dam site	Year of Installation	Working Status	Period for which data is available
1.	Water Level Sensor	3	1	2024	Y	
2.	Plumb Bob	-	Nil			
3.	Inclinometer	-	Nil			
4.	Stress meters	-	Nil			
5.	Strain meters	-	Nil			
6.	Toe Drain	Yes	Nil			
7.	Drain Wells	Yes	Nil			
8.	V-Notches	Yes	Nil			
9.	Pressure Gauges	-	Nil			
10.	Accelerograph	-	Nil			
11.	SCADA	-	Nil			
12.	Surveillance	C.C. Cameras	Nil			
13.	Rain Gauge ORG	Yes	10	Not Known	Y	
14.	Rain Gauge SRRG	Yes	3	1920	Y	

5.8.2 Data storage, analysis, and dissemination methods.

Collected rain gauge data is analyzed for the reservoir operations.

5.8.3 Recommendation on Instrumentation for dam health

As per the requirements of this dam, the existing instruments are sufficient.

Chapter 6: Structural and Geotechnical Evaluation

6.1 Review of Seismic parameters

This parameter will be reviewed at the time of detailed review of the dam.

6.2 Detailed assessment of dam stability

As the Dam is over 100 years old, the detailed design of the dam is not available. However, it is recommended to carry out detailed stability analysis of dam and its appurtenant works to be assess its stability.

6.3 Review of design drawings and structural modifications.

No original design drawings available. There is it is recommended to carryout the detailed design review as per the current codes and seismic parameters.

6.4 Recommendations based on latest codal provisions

As recommended above.

Chapter 7: Hydrologic and Hydraulic Evaluation of the Dam

A. Hydrological Evaluation:

7.1 Review and verification of the inflow design flood (IDF) and spillway capacity.

As per the Hydrology review carried out by the CWC, the dam qualifies for PMF. As per the analysis done the inflow value worked out as 6137 Cumec.

7.2 Assessment of downstream hazard based on population at risk and inundation mapping.

The surplussing capacity of the existing arrangements is far less than the inflows expected in the reservoir. Therefore, the downstream population is under great risk . As inundation mapping is not carried out, the extend of the hazard can't be be quantified.

7.3 Review of dam breach analysis and inundation maps, including assumptions and limitations.

The review of dam breach analysis and inundation maps, including assumptions and limitations has not been carried out so far. This analysis needs to be carried out on urgent basis

7.4 Suggest improvements or updates to mapping if outdated.

No mapping has been done. While preparing the inundation plan for the downstream area, the inundation plan for the upstream areas also to be carried upto the revised Maximum Water Level obtained after the review of the hydrological design.

B. Hydraulic Evaluation:

The downstream apron below the spillways consists of natural sheet rock and no apparent distress is seen, inspite of many years of operation of the spillway.

7.5 Sedimentation Survey :

7.5.1 According to the latest Hydro-Graphic Survey of Osman Sagar, the loss of storage in the gross capacity of the reservoir is worked out to be 81.0M.cum or 45 % over a period of 104 years. The loss of the capacity is quite substantial, and the same would grossly increase the downstream impact of floods and also reduce the volume of water available to fulfill the drinking water requirements of the City.

7.5.2

In as much as increasing the existing Reservoir capacity is beneficial for drinking water and flood control, Irrigation Department and HMWSSB may explore the feasibility of desilting Osman Sagar. A similar requirement be explored for Himayat Sagar also**.**

Chapter 8: Dam Safety Plans

8.1 Review of design documentation

Sl No	Particular	Last review Date	Current review date	Remarks
1	Review of Design Inflow	2020	To be carried in 2030	
2	Review of EAP	Not prepared	--	
3	Review of O & M	Prepared in 1990	Review yet to be carried as per the current guidelines.	
4	Review of Free board	--	Shall be reviewed at the time of structural design of the dam.	
5	Review of Rule curve	Not prepared	To be prepared	
6	Review of reservoir water quality	Every Month	01-11-2025	

8.2 Recommendation on Early Warning System (EWS)

8.2.1 Method of Inflow forecasting

At present, the rainfall gauges data received through Telangana State Development and Planning Society (TGDPS), Doppler Radar images through IMD and estimating runoff for the reservoir gate operations. It is observed from the location of the raingauges installed in the catchment area, there is a huge area where no raingauges are provided by TGDPS.

It is recommended that additional automatic raingauges be installed in these areas also to improve the quality of forecasting.

In addition, it is recommended that automatic GPRS based stream gauges be installed along the course of the river to improve the advance warning and quality of forecasting.

8.2.2 Method of downstream population alert

At present, the following methodology is adopted.

1. Activation of Siren installed at the top of the dam
2. Informing the Police, Revenue (RDO and MRO), Irrigation, Municipal, GHMC Control room and Telangana integrated Command Control Room (TGiCCC) authorities, through mobile phone calls and also being hosted in the whatsapp group

"Musi River Alert Team", 2 hours in advance before opening of the gates initially and 1 hour prior to for any increment, for informing public at large.

3. In addition to the above, information is being passed on to the electronic media through Public Relation Officer, HMWSSB, every hour.

The project authorities informed that the above arrangements are working well.

It is recommend that

1. Emergency action plan is to be prepared immediately and inundation maps are to prepared for maximum flood and areas of inundation during floods are to be disseminated to all the concerned stake holders including downstream affected population and awareness is to be created regarding intensity, spreaded area and effectiveness.
2. Evacuation routes and spouts on the downstream of the dam are to be identified and made known to the concerned authorities and stake holders.
3. Stake holder consultation meetings are to conducted with the population at downstream affected areas before every monsoon to get them awareness on the emergency situations.
4. Mock drills are to be conducted with stake holders.

Chapter 9: Detailed Safety and Design Evaluation

Summary of inspection of different components, instrumentation data and analysis performed.

1. **Back ground:**

   The catastrophic floods to Musi River during September 1908, has resulted in substantial loss of lives and property in Hyderabad. The twin reservoirs of Osman Sagar and Himayat Sagar were constructed to mitigate the affects of flood, and also to fulfill drinking water to the twin Cities, Irrigation requirement.

   a. When heavy inflows were received a quantity of 378 cumec was released from 9.00 pm on 26/11 to 5 am on 27/11.

This flood is hardly 15 % of the release capacity of the Spillway of Osman Sagar, but has caused serious disruption to the service Road and the Ring Road, existing close to the Reservoir. Further, disorder along the stream course down below in Hyderabad City was also reported. The visual depicts the rise of water level by 1-1.5m at the first of the two causeways existing in the Musa River Bed. Fundamental requirement of a flood channel is to design bedfall and surface fall based on Conservation of Mass and Energy. The back water curve induced by the said afflux makes the flows sluggish, leading to silting of the River Bed, and inundation of adjoining areas. This condition appears like lack of co-ordination among the line departments to fulfill the Hydrologic needs of the River. It was also reported that free flow of Musi is similarly constricted and in the reaches existing down below.

b. Surplus capacity from the Reservoir

Existing Spillway

For an inflow quantity of 2773.34 cumec, four types of surplus arrangements were provided for the Osman Sagar project i.e; flood gates, two sets of open vents and three numbers of By-wash arrangements, to release a total outflow of 2607 cumecs. However the Bywash structures are not traceable on the ground, and the current release capacity available is only 1554 Cumecs (only 60% from design value). As per the latest Hydrograhic Survey, the reservoir has also lost 45%of the gross storage capacity from inception. For such a large quantity of loss, the inherent ability of the Reservoir to delay and moderate the flood, stands reduced substantially.

c. Requirement of additional spillway

In as much as the dam qualifies for adoption PMF (BIS Code-11223-1980) the CWC have suggested to adopt 6137 cumecs. This cannot brushed aside stating to be too Hypothetical, as the Rainfall adopted in developing this Hydrograph has already occurred in some parts of the State of Telengana, especially during the period last one decade. The designed inflow of PMF needs to be routed through the existing condition of the to arrive at the Design outflow. Thus, it is necessary to provide the additional surplus arrangements from the dam. The PoE has observed that a good quality hard granite rock is available at shallow depth in the alignment of the Dam where it crosses the river course for the location of additional spillway. A feasibility study in this regard needs to be carried out immediately duly considering all the available options.

2. Spillway body material in the gated and gated Sections is in Stone Masonry and the same is Observed to be in good condition. Energy Dissipation Arrangement is generally found satisfactory. The scour that has formed adjoining the cistern on the left side needs to be filled up as suggested.
3. Gate Hoisting Arrangements

   The, pulleys, shafts, bush bearings need to be checked and lubricated. The rope drum, shafts, supporting plates, bolts, clamps to be checked, for deviations. Rusting and pitting to be cleaned on hoisting support structure. For the hoist bridge members, gantry rails, chequered plates, railings are to be periodically checked for corrosion, rusting and pitting periodically. The water stagnation over the hoist bridge to be avoided. The weak ropes are to be replaced. The bulk head gate to be lowered to attend main gate repairs replacements and maintenance.

4. Gates, seals and embedded parts:
   The skin plate and skin plate joints to be checked for corrosion and sections to be made-up suitably with welding and then painted. The gates and its structural components need to be painted. Seals to be checked and replaced wherever necessary. The rollers are to be cleaned and checked for smooth functioning. Water stagnate on web sections should not take place. During the dry sill condition (or) by lowering bulk head gate, the gates are to be inspected in detail.

5. Composite NOF section
   a. Masonry portion
      The masonry is generally in good condition, however the seepage marks are observed at some location because of rainwater falling on the dam top road and flowing down through the joint of the road and parapet wall.

   b. Earthen Portion

      The earth slope is in very bad condition covered with thick vegetation and trees. The slope at many places having undulations and not as per design. Chute drains are either damaged or missing. Rock toe and toe drains are full of debris. Dam face is covered with full vegetation.

6. Detailed assessment of the Dam

The Dam is over 100 years old, the detailed design of the dam are not available. However, it is recommended to carry out detailed stability analysis of dam and its appurtenant works to be assess its stability.

7. Design Review

No original design drawings available. The design review needs to be carried out as per the current codes and seismic parameters.

8. Early warning System

ARG stations need to be established in the vast uncovered area in the catchment. GPRS based stream gauges be installed along the course of the river to improve the advance warning and quality of forecasting.

Main Spillway:

Flood Gates - 15 nos(R.L 540.74) from Ch.no – 44 to 46

Auxiliary Spillway:

1. Open Vents – 15 nos (R.L 546.65) from Ch.no – 44 to 46
2. Open Vents -15 nos (R.L 545.59) from Ch.no – 58 to 60
3. Natural Bye Wash– 3 nos (R.L 550.20) location is not traceable.

9.1.1.1 This scoured area to be filled with high performance concrete (M30). This concrete to be provided with anchors 20mm dia, 1 mtr long both towards bed rock and the concrete @ 1.5m C/c in three rows in staggered manner.

9.1.1.2 Recommendations on the Gate Hoisting Arrangement, if any.

1. The pulleys positions with respect to cheek plates need to be checked along with shafts, bush bearings etc. Any deviations, if found, are to be attended. Lubrication is be carried out periodically.

2. The rope drum, drum shafts, supporting plates, tightness and sound ness of bolts and rope clamps to be checked, deviations to be attended.

3. Drum gear bolts to be checked for tightness and soundness. The mis alignments of gears to be checked and adjusted. Ensure uniform meshing of teeth.

4. On the hoist supporting structure, rusting and pitting to be cleaned. The structural members are to be checked.The weak members, if found, are to be strengthened or replaced. Foundation bolts to be checked.The damaged bolts, if found, to be replaced.

5. The hoist bridge members, gantry rails, chequered plates, railings are to be checked for corrosion, rusting and pitting periodically. The weak sections, if found, are to be repaired/replaced. The water stagnation over the hoist bridge to be avoided.

6. The braking arrangements are to be checked for intermediate stops.

7. The weak ropes are to be replaced.

9.1.1.3 Recommendations on the Gantry crane Arrangement, if any.

It is reported that the bulk head gate have been repaired. However, it could not be checked by lowering due to the reason that the crane was not operational because it is under repair. The bulk head gate to be lowered to attend main gate repairs replacements and maintenance.

Lifting beam

The lifting beam is not use since a long period.

9.1.1.4 Observations on the Gates, seals and embedded parts

1. This year the gates are operated more number of times during floods compared to past few years. The field staff reported that the maximum flood gate opening in the last 60 years occurred on 26.09.2025 from 9.00 AM to 8.00 PM and on the following day 15gates were opened to a height of 9 Feet. The moving components might have got affected due to these operations and heavy flood discharges.
2. These gates over 100 years old. The gates are with corrosion, rusting and pitting sparsely.
3. The skin plate, skin plate joints, joints between skin plates and horizontal girders, vertical stiffeners in wet condition if exposed to atmosphere are subjected to corrosion and rusting. The foreign materials, particularly the sharp edge debris, uneven surface of seal seat, uneven loads may cause damages. The bottom seal damages due to uneven contact of seal and seal seat, loose fixing bolts. The seals might be affected.
4. The roller train supporting plates, lock plates, rollers, pins, bushes will get affected due to uneven forces. The rollers assemblies might have been affected. The rollers without maintenance will get jammed.
5. The horizontal girder with water stagnation, in wet condition exposed to atmosphere causes rusting and pitting.
6. The metal housing embedded, the roller path, seal seat, pad seat might have affected due to accumulation of foreign material, debris, trash.
7. The metal housing (groove) facilitate roller movement, seal seat, pad seat might have been affected heavy floods, more operation and being old.

9.1.1.5 Observations on the Dam body material

1. The stone masonry is in good condition.
2. The earthen dam portion is fully covered with vegetation and trees and therefore could not be examined thoroughly. As per the dam authorities, no leakages/seepages observed through the body of the dam.

9.1.2 Downstream slope:

9.1.2.1 Observations on the Downstream slope

C. Earthen portion:

1. The earth slope is in very bad condition covered with thick vegetation and trees.
2. The slope at many places having tree stubs needs to be removed along with its roots.
3. The slope is not maintained as per the original design.
4. The chute drains are either damaged or missing.
5. Dam rock toe and toe drain are fully covered with debris.
6. No instrumentation is provided.

7. As the face is covered fully with vegetation, its condition could not be evaluated for any seepage taking place. However, no flowing water was observed.

D. Masonry portion:

The masonry is general in good condition, however the seepage marks are observed at some location because of rainwater falling on the dam top road and flowing down through the joint of the road and parapet wall.

9.1.2.2 Recommendations on the Downstream slope, if any.

A. Earthen Portion:

1. The vegetation to be cut and uprooted including stubs of the trees.
2. The trees to be cut and uprooted after obtaining permission from Forest Department
3. After carrying out the review of the stability, the earthen slope to be brought to the design section.
4. Damaged chutes to be repaired/re constructed.
5. Rock toe including toe drain be cleared off debris and made functional.
6. V-Notches to be provided at suitable locations for the measurement of seepage water.

Piezo meters to be

9.1.3 Downstream toe drain:

9.1.3.1 Observations on the toe drain

1. Dam rock toe and toe drain are fully covered with debris.
2. Due to above reason no observation of seepage water are being carried out.
3. Due to the same reason, there is no access to the toe area.

9.1.3.2 Repercussions, if the issue is not addressed, if any.

If the Toe drains are not operational, there would be build up of pore water pressure leading to instability of the slope.

9.1.3.3 Recommendations on the toe drain, if any.

The rock toe and toe drain to be cleared off all the debris and made operational.

9.1.4 Recommendation on Possibility of Failure

9.2 Detailed assessment of dam stability

As the Dam is over 100 years old, the detailed design of the dam is not available. However, it is recommended to carry out detailed stability analysis of dam and its appurtenant works to be assess its stability.

No original design drawings available. There is it is recommended to carryout the detailed design review as per the current codes and seismic parameters.

As per the Hydrology review carried out by the CWC, the dam qualifies for PMF. As per the analysis done the inflow value worked out as 6137 Cumec.

The surplussing capacity of the existing arrangements is far less than the inflows expected in the reservoir. Therefore, the downstream population is under great risk . As inundation mapping is not carried out, the extend of the hazard can’t be be quantified.

The review of dam breach analysis and inundation maps, including assumptions and limitations has not been carried out so far. This analysis needs to be carried out on urgent basis.

9.2.1 According to the latest Hydro-Graphic Survey of Osman Sagar, the loss of storage in the gross capacity of the reservoir is worked out to be 81.0M.cum or 45 % over a period of 104 years. The loss of the capacity is quite substantial, and the same would grossly increase the downstream impact of floods and also reduce the volume of water available to fulfill the drinking water requirements of the City.

Chapter 10: Recommendations and Action Plan

10.1 Recommendation on Immediate threat to dam structural safety, if any-

A. Gross inadequacy in the surplus arrangement existing in the project:

1. In the year 1908, a severe flood event was experienced in the river Musi catchment which resulted in a loss of about 15 thousand lives out of the total population of about 5 lakhs at that time. This event prompted the Nizam government to think of having a flood management structure in the catchment which could also be used for supply of drinking water to the inhabitants to the city of Hyderabad.

2. In the original design the estimated maximum discharge at the Dam site was considered as 2973 Cumecs. Accordingly, the dam was provided with the outflow capacity of 2608 Cumecs. During the field visit of the PoE of Telangana State to the project on 13.11.2025, it was found that the natural bye wash arrangement to dispose 1053 Cumecs is currently not available. Hence,1555 Cumec only is actually available to get discharged through the balance surplussing arrangement existing at the project site.

3. In the latest hydrological review carried out for the project in 2020, the design inflow in to the Osman sagar reservoir is estimated to be of the order of 6137 Cumecs. This indicates that there is gross inadequacy in the surplussing arrangements presently existing at the dam site.

4. Keeping the above circumstances, it is the opinion that the safety of dam is in serious risk and necessary action to enhance the surplussing arrangements needs to be taken up urgently.

5. Compounding to the above-mentioned problem, the carrying capacity of the river Musa & the river Musi has also over the year got reduced because of siltation, waste materials dumping and uncontrolled encroachments. This deficiency came to light on Dt 26& 27th September 2025 when even with a small discharge of 377 Cumecs in the river Musa and combined discharge of 985 Cumecs in the Musi River which included the discharge of River Esa created inundation in some downstream areas.

B. Heavy vegetation including large size trees existing on the downstream earthen section of the composite dam:

1. The surface of an earthen dam has to be clear of vegetation to enable the maintenance staff to periodically inspect the dam surface for any leakage, slips, bore holes etc., which may disturb the slope on both the sides.

2. During the inspection by the PoE on 13-11-2025, it was observed that the earthen portion of the composite dam, on both the sides, are completely covered with thick vegetation which include large size trees. Such thick vegetation will hinder the earthen section from providing the passive resistance considered in the design. Further, this vegetation cover prevents inspection of the earthen surface of any distress. In addition, the roots of the

trees damage the interior of the earthen dam body and is likely to create piping etc., resulting into leakages and thereby the stability of the earthen section of the dam.

C. Design review of the project:

1. The dam was designed in the year 1912 based on the prevailing design practices of that period. Subsequently the BIS codes were formulated with a standardized procedure. Special provisions are also developed taking earthquake coefficients. Accordingly, design of all the dam components needs to be reviewed for structural adequacy and soundness as per the current practices and codes. This design review is also mandatory as per the Dam Safety Act-2021.

2. The rule curve for the reservoir operation needs to be prepared for safe management of the floods likely to be received in the reservoir, as per the reviewed hydrological analysis.

D. Re-sectioning of the rivers Musa, Esa and Musi :

As brought out above underPara 10.1.A 5 above, the discharging capacity of Musa, Esa & Musi rivers over the years has got reduced due to siltation, debris and encroachments. This is far less than design discharges of Osman sagar and Himayathsagar dam projects, which are separately estimated to be 6137 Cumecs for Osmasnagar and 12,035 Cumecs (yet to be reviewed) for Himayathsagar. Since the origins of the two rivers are from the same Ananthagiri Hills range, their catchment areas have common boundary and thus may share a storm of large magnitude may occur simultaneously in adjoining catchments. This may result in a combined flow may much higher than 4247 Cumec (1.50 Lac Cusecs) recommended by expert committee constituted vide GO Ms No.61 dated: 01-05-2012, report submitted in 2017. As the latest PMF studies shown much higher values in both the catchments, the situation is more alarming.

In case, a storm event as enumerated above happens, the City of Hyderabad is going to face a very severe flood condition, inundating large areas of the city and also heavy loss of life, property and valuable assets.

10.2 Recommendation on further studies for dam safety

1. As the current spilling capacity is grossly inadequate as cited in the above paras, it is necessary to create additional surplussing arrangement to enhance the capacity. During the field inspection by the PoE it was observed that a good quality of hard granite rock is available at shallow depth in the alignment of the Dam where it crosses the river course. Prime a facia it appears to be a good location which could be considered for the location of the additional spillway for surplussing balance flood discharge. A feasibility study in this regard needs to be carried out immediately. In addition to this, any other suitable location for locating a similar structure may also to be explored.

2. The L-Section and Cross section survey of rivers Musa and Esa upto the confluence and the river Musi upto atleast 5km downstream, where it exits Hyderabad city area to evaluate their present carrying capacity and thereafter re-sectioning of the river cross

sections will be required.If required, a protection in the form of Earthern bund or a retaining wall along the course of the river may have to be considered for the protection of the Hyderabad City.

3. A mathematical model study of this river system is to be carried out to establish the revised flood levels enroute. Any river development along these rivers to be carried out keeping in view these revised established high flood levels.

4. A rainfall runoff model needs to be developed to help in the optimum operation of the reservoir.
5. Dam Break Analysis to be carried out and inundation plans to be prepared accordingly to demarcate the highly vulnerable areas which may severely get affected by a flood of High magnitude.
6. Emergency Action Plans be prepared for any emergency that could get created because of High Flood situation.
7. The rule curve for the reservoir operation needs to be prepared for safe management of the floods likely to be received in the reservoir, as per the reviewed hydrological analysis.

HYDRO MECHANICAL:

1. The hydro mechanical equipment are 100 years old.Therefore, Non Destructive Testing (NDT) of the gate components like Skin Plates, Vertical Stiffener, Horizontal Girders, End Girders, of gates are to be conducted on priority.

   N D T shall be conducted as per Indian Standard codes .

   The test results are to be verified with design strength requirement. The decision may be taken for repair/replacement on the basis of the NDT test and as per the advice of the metallurgy experts.

Gantry crane:

The gantry crane repairs are taken up. The Project Authorities sated that as crane is manually operated, bulkhead gate operation is taking more time. Power operation of the gantry cranewill reduce the operation time considerably. The existing speed reduction system is of manual operation is to be modified suitably to suit power operation.

10.3 Recommendation on Operation schedule and maintenance schedule

1. An operation and maintenance schedule which is available for reference by the hydromechanical staff engaged in the operation and maintenance of the hydromechanical works is Old and need to be updated as per the Dam Safet Act. The pre-monsoon inspection reports, post monsoon inspection, periodical inspection reports, details recorded in operation and maintenance registers, the breakdowns took place during

operation are to be recoded systematically.These will indicate the health status of the equipment.

It is observed that the operation and maintenance staff available at site are inadequate, this may affect operation and maintenance of hydro mechanical equipment. The trained and experienced personnel may be deployed to have a smooth functioning of the system.

2. Site specific seismic parameters to be evaluated and used for design review of the gate and its appurtenances.

10.4 Recommendation on major rehabilitation workfor dam structural safety, if any

1. The safety of dam is in serious risk and necessary action to enhance the surplussing arrangements needs to be taken up urgently.
2. The surface of an earthen dam has to be clear of vegetation to enable to maintenance staff to periodically inspect the dam surface for any leakage, slips, bore holes etc., may have operation of the reservoir.
3. The design of all the dam components needs to be reviewed for structural adequacy and soundness as per the current practices, codes and relevant earthquake coefficients applicable to the site. This design review is mandatory as per the Dam Safety Act-2021.

   After the detailed design review of the components of the dam, the rehabilitation work needs to be taken as required.

4. A Rule Curve for the reservoir operation needs to be prepared for safe management of the revised design flood, duly considering the requirements of dam safety, drinking water requirements of Hyderabad city and downstream flooding.
5. The extreme events have become frequent in Telangana state, significantly, in the last one decade. The sections of rivers Esa, Musa & Musi need to be modified wherever necessary to accommodate the resultant flows arising due to the revised inflows by way of PMF at Osman Sagar and Himayath Sagar reservoirs. Such modifications are to be carried out up toat least 5km downstream, where the river Musi exits Hyderabad city.
6. The dam was designed for an inflow of 2973.34Cumecs and corresponding outflow capacity of 2607.98 Cumecsis provided. Currently only 1555 Cumecs of surplus capacity is available. The revised inflow for the project PMF is worked out as 6137 Cumecs by CWC. The corresponding outflow capacity needs to be computed for the revised inflow of the project. Thus, it is necessary to provide the additional surplus arrangements from the dam.
7. The PoE has observed that a good quality hard granite rock is available at shallow depth in the alignment of the Dam where it crosses the river course for the location of additional spillway. A feasibility study in this regard needs to be carried out immediately duly considering all the available options.
8. A mathematical model study of this river system is to be carried out to establish the revised flood levels enroute. Any river development along these rivers to be carried out keeping in view these revised established high flood levels.
9. The Hydro-mechanical equipment has become old and need to be verified for metal condition via N D T tests as per BIS standards andby a qualified metallurgist and necessary modifications may be carried out accordingly.

10. Necessary improvements may be carried out for power operation of the Gantry Crane.
11. The Original drawings with reference to the various components of the project by the way of plans and cross sections are reported to be not available. This study was carried based on Google Map, visible features on the ground and corresponding inferences were made. It is necessary to get all the as built drawings prepared for taking decisions suggested in the report. It is also necessary to arrange to prepare a complete Layout drawingincluding various components of the project.

10.5 Include suggestions for next evaluation period

1. In the current dam safety evaluation, various dam safety recommendations to be carried out were mentioned in therefore paras. These recommendations are to be carried out immediately. It is necessary to verify whether the recommendation were carried out expeditiously without further loss of time.

Annexures

GOVERNMENT OF TELANGANA
IRRIGATION AND CAD (REFORMS) DEPARTMENT

Memo.No.6387/Reforms/A1/2022 **Dated: 31.10.2022**

Sub: Irrigation & CAD Department – Chief Engineer – State Dam Safety Organization – Periodical and Comprehensive evaluation of the of the Large Dams (NRLD) – Permission for Constitution of Panel of Experts - Orders – Issued.

Ref: 1. Dam Safety Act-2021.
2. From the EnC(G), Hyd., Lr.No.ENC(G)SE(P&M)/DCE(GB)/ DEE1 /AEE3/Dam Safety/PoE Committee, Dt:27.09.2022 and Dt.24.10.2022.

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Under the circumstances reported by the Engineer-in-Chief (General), I&CAD Department in the reference 2nd cited, and in terms of Section 38 (1) of the Dam Safety Act of 2021, Government after careful examination of the matter hereby Constitute the Panel of Experts (PoE) with the following members to conduct periodical and comprehensive evaluation of the Large Dams (NRLD).

Sl No.	Name of the Officer	Expertise in	Designated as
1	Sri Ashok Kumar Ganju, Member (Design & Research), Retd. CWC , New Delhi	Dam Safety	Chairman
2	Sri M. Raju, Director GSI (Retd)	Geology/ foundations	Member
3	Sri B.Muralidhar Chief Research Officer / Scientist 'D' (Retd), CWPRS	Geotechnical/ Instrumentation	Member
4	Sri Dr. M.V.S. Sreedhar Head of Geo- Technical Dept., Osmania University	Geotechnical	Member
5	Sri K.Satyanarayana Chief Engineer (Retd)	Mechanical	Member
6	Sri T. Khagendar Chief Engineer (Retd)	Mechanical	Member
7	Sri KS.S Chandra Shekar Working Chief Engineer, O/o Engineer-in-Chief (G), I & CAD Dept. Telangana.	Civil	Member & special Invitee
8	Sri T.Rajasekhar Superintending Engineer (Retd)	Civil	Member
9	Sri C.Mahender Superintending Engineer(Retd)	Civil	Member
10	Sri Y.V.Krishna Rao Deputy Executive Engineer (Retd.) CDO	Mechanical	Member

2. The remuneration for the above Panel of Experts with permission to hire 2 Nos. of Innova Vehicles (during inspections only) is as follows:-

1. The remuneration for Chairman Rs.12,000/- + GST per day and shall not exceed Rs.1,20,000/- per month.
2. The remuneration for members Rs.10,000/- + GST per day and shall not exceed Rs.1,00,000/- per month.

(Cntd.Page.2)

3. A fixed amount of Rs.800/- will be paid towards conveyance expenses on the day of service utilization at Hyderabad.

4. Payment will be made for Air/Train Travel as follows:
   i. Economy class by any airlines for any journey by air.
   ii. First class AC for train Travel.

5. For outstation journey, they will be paid for Boarding / Lodging and other travelling expenses as per actual.

3. The above expenditure shall be debited from Head of account 2700-80-800-25-52-280-284.

4. The Engineer-in-Chief (General), Irrigation & CAD Department, Hyderabad shall take further necessary action, accordingly.

Dr. RAJAT KUMAR
SPECIAL CHIEF SECRETARY TO GOVERNMENT

To
The Engineer-in-Chief (General), I&CAD Department,
Jalasoudha Buildings, Errum Munzil, Hyderabad .
The Chief Engineer, CDO, I&CAD Department, Hyderabad.

Copy to:
The Accountant General, Telangana, Hyderabad.
The P.S. to Spl. Chief Secretary to Govt. I&CAD Dept.,
Sc/Sf.

//FORWARDED::BY ORDER//

P.v.v.s.[illegible]

SECTION OFFICER

GOVERNMENT OF TELANGANA
IRRIGATION AND CAD (REFORMS) DEPARTMENT

Memo.No.7898/Reforms/A1/2022 **Dated: 09.12.2022**

Sub: I & CAD Dept - Chief Engineer- State Dam Safety Organization – Periodical and Comprehensive evaluation of the Large dams (NRLD)–Inclusion of Sri P.Rama Raju, Engineer-in-Chief (Retd.) as a Member of Panel of Experts in the already constituted POE – Regarding.

Ref: 1.Govt.Memo.6387/Reforms/A1/2022, Dated.31.10.2022.
2.From the EnC(G), Hyd., Lr.No.ENC(G)SE(P&M)/DCE(GB)/ DEE1 /AEE3/Dam Safety/PoE Committee, Dt:04.12.2022.

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Under the circumstances reported by the Engineer-in-Chief (General), I&CAD Department in the reference 2nd cited, Government after careful examination of the matter, hereby include Sri P.Rama Raju, Engineer-in-Chief (Retd.) as a Member in the Panel of Experts (NRLD) for Periodical and Comprehensive evaluation of the Large Dams as per Dam Safety Act 2021, with the same terms and conditions stipulated in the reference 1st cited.

2. The Engineer-in-Chief (General), Irrigation & CAD Department, Hyderabad shall take further necessary action, accordingly.

Dr. RAJAT KUMAR
SPECIAL CHIEF SECRETARY TO GOVERNMENT

To
The Engineer-in-Chief (General), I&CAD Department,
Jalasoudha Buildings, Errum Munzil, Hyderabad .
The Chief Engineer, CDO, I&CAD Department, Hyderabad.
Copy to:
The Accountant General, Telangana, Hyderabad.
The P.S. to Spl. Chief Secretary to Govt. I&CAD Dept.,
Sc/Sf.

//FORWARDED::BY ORDER//

P.V.V.S. Seshakumar

SECTION OFFICER

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अधिसूचना

नई दिल्ली, 23 मई, 2025

फा. सं. टीई-32/2/2023-एनडीएसए-एमओडबल्यूआर.—राष्ट्रीय बांध सुरक्षा प्राधिकरण, बांध सुरक्षा अधिनियम, 2021 (2021 का 41) की धारा 38 की उपधारा (1) के साथ पठित धारा 54 की उपधारा (1) और उपधारा (2) के खंड (न) द्वारा प्रदत्त शक्तियों का प्रयोग करते हुए, राष्ट्रीय बांध सुरक्षा समिति की सिफारिशों पर, निम्नलिखित विनियम बनाता है, अर्थात: -

1. **संक्षिप्त नाम और प्रारंभ** – (1) इन विनियमों का संक्षिप्त नाम प्रत्येक विनिर्दिष्ट बांध विनियम, 2025 के व्यापक बांध सुरक्षा मूल्यांकन के लिए विशेषज्ञों का स्वतंत्र पैनल है

   (2) ये राजपत्र में उनके प्रकाशन की तारीख को प्रवृत्त होंगे।

2. परिभाषाएं - (1) इन विनियमों में, जब तक कि संदर्भ से अन्यथा अपेक्षित न हो, -

(क) "अधिनियम" से बांध सुरक्षा अधिनियम, 2021 (2021 का 41) अभिप्रेत है;

(ख) "प्राधिकरण" से अधिनियम की धारा 8 के अधीन स्थापित राष्ट्रीय बांध सुरक्षा प्राधिकरण अभिप्रेत है

(2) उन शब्दो और पदों के जो इसमें प्रयुक्त हैं, और इन विनियमों में परिभाषित नहीं हैं, किन्तु अधिनियम में परिभाषित हैं, वही अर्थ होंगे जो उस अधिनियम में हैं।

3. प्रत्येक विनिर्दिष्ट बांध के व्यापक बांध सुरक्षा मूल्यांकन के लिए विशेषज्ञ का स्वतंत्र पैनल — विनिर्दिष्ट बांध और उसके जलाशय की स्थिति अवधारित करने के प्रयोजन के लिए स्वतंत्र विशेषज्ञ का स्वतंत्र पैनल की संरचना निम्नानुसार है: —

(1) विशेषज्ञों के स्वतंत्र पैनल में प्रमुख विशेषज्ञ जैसे अध्यक्ष और सदस्य सम्मिलित होंगे।

(2) पैनल में प्रमुख विशेषज्ञों की संख्या बांध सहयुक्त पहचाने गए या ज्ञात बांध सुरक्षा वाद-विषय और परिसंकट के आधार पर पहचान की जाएगी, तथापि किसी दशा में विशेषज्ञों के स्वतंत्र पैनल में तीन से कम प्रमुख विशेषज्ञ नहीं होंगे, जिसमें बांध सुरक्षा विशेषज्ञ की उपस्थिति अनिवार्य होगी तथा पैनल के विशेषज्ञों के लिए अपेक्षित अर्हताएं और कौशलों के ब्यौरे सारणी – 1 में निर्दिष्ट है।

(3) स्वतंत्र पैनल के अध्यक्ष और सदस्यों में केंद्रीय सरकार संगठनों, राज्य सरकारों, पब्लिक सेक्टर के उपक्रमों, भारतीय प्रौद्योगिकी संस्थान/राष्ट्रीय प्रौद्योगिकी संस्थान, ख्यातिप्राप्त प्राइवेट संस्थाओ, ख्यातिप्राप्त स्वतंत्र विशेषज्ञों के कार्यरत के साथ सेवानिवृत्त अधिकारी सम्मिलित हो सकेंगे।

(4) विशेषज्ञों के स्वतंत्र पैनल के अध्यक्ष और सदस्य हितों के टकराव जिसके अंतर्गत वित्तीय, व्यावसायिक या व्यक्तिगत संबंध भी है से मुक्त होंगे, जो उनके निर्णय को प्रभावित कर सकते हैं और यह सुनिश्चित किया जाएगा कि पैनल के सदस्यों का उनके कार्य से प्रभावित इकाइयों या हितों के साथ कोई प्रत्यक्ष संबंध न हो।

(5) अंतरराज्यीय विनिर्दिष्ट बांधों से भिन्न प्रत्येक विनिर्दिष्ट बांध स्वामी, जिनके लिए प्राधिकरण राज्य बांध सुरक्षा संगठन है, विशेषज्ञों का स्वतंत्र पैनल गठित करेगा और अंतरराज्यीय विनिर्दिष्ट बांधों की दशा में, जिनके लिए प्राधिकरण राज्य बांध सुरक्षा संगठन है, संबंधित राज्य बांध सुरक्षा संगठन का प्रमुख पैनल गठित करेगा।

(6) विशेषज्ञों को उनकी अर्हताएं और विशेषज्ञता के संबंधित क्षेत्र में अनुभव तथा उनके व्यवसाय में प्रतिष्ठा के आधार पर वर्गीकृत किया जाएगा, जैसा कि सारणी- 1 में दिया गया है।

सारणी-1

क्रम संख्या	विशेषज्ञ	न्यूनतम अर्हता	अनुभव
(1)	(2)	(3)	(4)
1.	बांध सुरक्षा विशेषज्ञ	सिविल इंजीनियरिंग में बैचलर ऑफ इंजीनियरिंग या बैचलर ऑफ टेक्नोलॉजी या समतुल्य डिग्री, अधिमानतः भू-तकनीकी इंजीनियरिंग या भूवैज्ञानिकी इंजीनियरिंग में मास्टर डिग्री या जल संसाधन इंजीनियरिंग या जल विज्ञान या भूविज्ञान या इंजीनियरिंग भूविज्ञान में मास्टर डिग्री	• कम से कम पांच बड़ी बांध परियोजनाओं के बांध सुरक्षा से संबंधित वाद-विषयों को संभालने का अनुभव यदि अभ्यर्थी राष्ट्रीय और अंतर्राष्ट्रीय स्तर पर किसी भी प्रकार की आपदाओं (विशेष रूप से बांध आपदाओं से संबंधित) से निपटने वाली टीम का सदस्य रहा हो तो उसे अतिरिक्त वरीयता दी जाएगी। • बांधों के डिजाइन, संनिर्माण, सुरक्षा मूल्यांकन या प्रचालन में कम से कम दस वर्ष का संबंधित अनुभव होना चाहिए। पुराने बांध संरचनाओं के पुनर्वास के अनुभव को अतिरिक्त वरीयता दी जाएगी। • बांधों के निरीक्षण, इंस्ट्रूमेंटेशन, आपातकालीन कार्रवाई योजना आदि का व्यावसायिक अनुभव।
2.	डिज़ाइन विशेषज्ञ	सिविल इंजीनियरिंग में बैचलर ऑफ इंजीनियरिंग या बैचलर ऑफ टेक्नोलॉजी या समतुल्य डिग्री, अधिमानतः	• कम से कम पांच बड़ी बांध परियोजनाओं की योजना तैयार करने और डिजाइन बनाने का अनुभव।

क्रम संख्या	विशेषज्ञ	न्यूनतम अर्हता	अनुभव
(1)	(2)	(3)	(4)
		संरचनात्मक इंजीनियरिंग या जलीय इंजीनियरिंग या जल संसाधन विकास में मास्टर डिग्री	• बड़ी बांध परियोजनाओं की योजना और डिजाइन में कम से कम दस वर्ष का संबंधित अनुभव होना। • पुराने बांध संरचनाओं के पुनर्वास के अनुभव को अतिरिक्त वरीयता दी जाएगी।
3.	संनिर्माण पर्यवेक्षण विशेषज्ञ	सिविल इंजीनियरिंग में बैचलर ऑफ इंजीनियरिंग या बैचलर ऑफ टेक्नोलॉजी या समतुल्य, अधिमानतः संरचनात्मक इंजीनियरिंग या जलीय इंजीनियरिंग या जल संसाधन विकास में मास्टर डिग्री	• न्यूनतम तीन विनिर्दिष्ट बांधों के निर्माण, कार्य पर्यवेक्षण और क्वालिटी आश्वासन का अनुभव। पुराने बांध ढांचों के पुनर्वास के अनुभव को अतिरिक्त वरीयता दी जाएगी। • बांध परियोजनाओं के संनिर्माण में कम से कम दस वर्ष का व्यावसायिक अनुभव।
4.	भूवैज्ञानिक	भूविज्ञान या इंजीनियरिंग भूविज्ञान में मास्टर डिग्री	• न्यूनतम पांच बड़े बांध परियोजनाओं के विभिन्न भूवैज्ञानिक पहलुओं को संभालने का अनुभव। • जल संसाधन परियोजनाओं के भूवैज्ञानिक इंजीनियरिंग पहलुओं से निपटने में कम से कम दस वर्ष का व्यावसायिक अनुभव
5.	जल-वैज्ञानिक	सिविल इंजीनियरिंग में बैचलर ऑफ इंजीनियरिंग या बैचलर ऑफ टेक्नोलॉजी या समतुल्य, अधिमानत: जल विज्ञान या जलीय इंजीनियरिंग या जल संसाधन विकास में मास्टर डिग्री	• प्रमुख बांध परियोजनाओं की जल विज्ञान संबंधी योजना बनाने में अनुभव, विशेष रूप से कम से कम दस बड़े बांध परियोजनाओं के बाढ़ आकलन, बाढ़ डिजाइन की समीक्षा, तूफान अंतर्गामी डिजाइन, अवसादन पहलुओं आदि में अनुभव। • विभिन्न सुसंगत जल विज्ञान प्रतिरूपण उपकरणों, बांध टूटने का प्रतिरूपण, जलप्लावन मानचित्रण, आपातकालीन कार्रवाई योजना के विकास और इसके कार्यान्वयन आदि के लिए किए गए कार्य का अनुभव रखने वाले विशेषज्ञ के लिए अतिरिक्त वरीयता। • जल संसाधन परियोजनाओं के जल विज्ञान संबंधी पहलुओं से निपटने में कम से कम दस वर्ष का व्यावसायिक अनुभव
6.	जल-यांत्रिक विशेषज्ञ	सिविल इंजीनियरिंग या यांत्रिक इंजीनियरिंग में बैचलर ऑफ इंजीनियरिंग या बैचलर ऑफ टेक्नोलॉजी या समतुल्य	• स्पिलवे द्वार, अंडर-स्लूस द्वार, आपातकालीन द्वार, जल-यांत्रिक कार्यों के निर्माण के पुनर्वास में अनुभव को अतिरिक्त वरीयता दी जाएगी। • पैनल में सम्मिलित होने से पहले पिछले दस वर्ष के दौरान सुसंगत अनुभव का प्रदर्शन

क्रम संख्या	विशेषज्ञ	न्यूनतम अर्हता	अनुभव
(1)	(2)	(3)	(4)
			• बड़े बांध परियोजनाओं के जल-यांत्रिक कार्यों के डिजाइन और विशेष रूप से विस्तृत इंजीनियरिंग परियोजनाओं और पुनर्वास जल-यांत्रिक कार्यों में कम से कम दस वर्ष का व्यावसायिक अनुभव।
7.	साधन- विनियोग विशेषज्ञ	बैचलर ऑफ इंजीनियरिंग या बैचलर ऑफ टेक्नोलॉजी, या सिविल इंजीनियरिंग या साधन विनियोग इंजीनियरिंग या इलेक्ट्रॉनिकी इंजीनियरिंग या समतुल्य या भौतिकी में मास्टर डिग्री	• न्यूनतम तीन बड़ी बांध परियोजनाओं के लिए सभी प्रकार के बांध उपकरणों की योजना और स्थापना का अनुभव। • पैनल में सम्मिलित होने के लिए आवेदन करने से पहले पिछले दस वर्ष के दौरान सुसंगत अनुभव का प्रदर्शन करें। पुराने बांध संरचनाओं के पुनर्निर्माण के अनुभव को अतिरिक्त वरीयता दी जाएगी। • बांध परियोजनाओं के डिजाइन में कम से कम दस वर्ष का व्यावसायिक अनुभव।
8.	भूकंप विशेषज्ञ	सिविल इंजीनियरिंग में बैचलर ऑफ इंजीनियरिंग या बैचलर ऑफ टेक्नोलॉजी या समतुल्य, अधिमानतः भूकंप इंजीनियरिंग या संरचनात्मक इंजीनियरिंग या भौतिकी में मास्टर डिग्री।	• भूकंपीय भार को ध्यान में रखते हुए न्यूनतम पांच बड़ी बांध परियोजनाओं की योजना और डिजाइन में अनुभव, सामयिक इतिहास श्रृंखला के साथ-साथ भूकंपीय स्पेक्ट्रा का अनुभव एक अतिरिक्त लाभ होगा और स्थल विनिर्दिष्ट भूकंपीय डिजाइन मापदंडों के लिए किए गए कार्य को अतिरिक्त वरीयता दी जाएगी। • पैनल में सम्मिलित होने के लिए आवेदन करने से पहले पिछले दस वर्ष के दौरान सुसंगत अनुभव का प्रदर्शन करें। पुराने बांध संरचनाओं के पुनर्वास के अनुभव को अतिरिक्त वरीयता दी जाएगी। • बांध परियोजनाओं के डिजाइन में कम से कम दस वर्ष का व्यावसायिक अनुभव।

अनिल जैन, अध्यक्ष

[विज्ञापन-III/4/असा./141/2025-26]

NATIONAL DAM SAFETY AUTHORITY

NOTIFICATION

New Delhi, the 23rd May, 2025

F. No.TE-32/2/2023-NDSA-MOWR.—In exercise of the powers conferred by sub-section (1) and clause (t) of sub-section (2) of section 54 read with sub-section (1) of section 38 of the Dam Safety Act, 2021 (41 of 2021), the National Dam Safety Authority, on the recommendations of the National Committee on Dam Safety, hereby makes the following regulations, namely: -

1. **Short title and commencement**. – (1) These regulations may be called the Independent Panel of Experts for Comprehensive Dam Safety Evaluation of each specified dam Regulations, 2025.

(2) They shall come into force on the date of their publication in the Official Gazette.

2. **Definitions.** - (1) In these regulations, unless the context otherwise requires, -

(a) "Act" means the Dam Safety Act, 2021(41 of 2021);

(b) "Authority" means the National Dam Safety Authority established under section 8 of the Act.

(2) Words and expressions used herein and not defined in these regulations but defined in the Act, shall have the meanings respectively, assigned to them in the Act.

3. **Independent panel of expert for comprehensive dam safety evaluation of each specified dam.**—The composition of Independent Panel of expert for the purpose of determining the condition of the specified dam and its reservoir are as follows: —

(1) The independent panel of experts shall consist of key experts- such as Chairperson and Members.

(2) The number of key experts in the panel shall be identified based on the identified or known dam safety issues and hazards associated with the dam, however in no case independent panel of experts shall consist of less than three key experts with mandatory presence of dam safety expert and the details of qualifications and skillsets required for the experts of the panel are elucidated in Table 1.

(3) The Chairperson and Members of independent panel of experts may comprise of working as well as retired officials from the Central Government Organisations, State Governments, Public Sector Undertakings, Indian Institute of Technology/National Institute of Technology, reputed private institutions, reputed freelance experts.

(4) The Chairperson and Members of independent panel of experts shall be free from conflicts of interest, including financial, professional, or personal ties that could bias their judgment and it shall be ensured that the panel members do not have any direct involvement with the entities or interests affected by their work.

(5) Each specified dam owners other than interstate specified dams for which Authority is the State Dam Safety Organisation, shall constitute an independent panel of experts and in case of the interstate specified dams for which Authority is the State Dam Safety Organisation, the head of the concerned State Dam Safety Organisation shall constitute the panel.

(6) The experts shall be classified on the basis of their qualifications and experience in the respective field of specialisation and the eminence in their profession as given in Table-1.

TABLE-1

Serial Number	**Experts**	**Minimum qualification**	**Experience**
(1)	(2)	(3)	(4)
1.	Dam Safety Expert	Bachelor of Engineering or Bachelor of Technology or equivalent degree in Civil Engineering Preferably Master Degree in Geotechnical Engineering or Geological Engineering Or Water Resources Engineering or Hydrology or Master degree in Geology or Engineering Geology	• Experience in handling dam safety related issues of minimum of five number of large dam projects • Additional weightage in case candidate had been a member of team that handled disasters of any type (especially related to dam disasters) at national and international level. • Having at least ten years of related experience in design, construction, safety evaluation, or operation of dams. Experience of rehabilitation of old dam structures would get additional weightage. • Professional experience of inspection of dams, instrumentation, emergency action planning etc.

Serial Number	**Experts**	**Minimum qualification**	**Experience**
(1)	(2)	(3)	(4)
2.	Design Expert	Bachelor of Engineering or Bachelor of Technology or equivalent degree in Civil Engineering Preferably Master Degree in Structural Engineering or Hydraulic Engineering or Water Resources Development	• Experience in planning and designs for minimum of five large dam projects. • Having at least ten years of related experience in planning and design of large dam projects. • Experience of rehabilitation of old dam structures would get additional weightage.
3.	Construction Supervision Expert	Bachelor of Engineering or Bachelor of Technology or equivalent in Civil Engineering Preferably Master Degree in Structural Engineering or Hydraulic Engineering or Water Resources Development	• Experience of Construction, Work Supervision and Quality Assurance for a minimum of three specified dams. • Experience of rehabilitation of old dam structures would get additional weightage. • Professional experience of at least ten years in construction of dam projects.
4.	Geologist	Master Degree in Geology or Engineering Geology	• Experience of handling various geological aspects of minimum of five large dam projects. • Professional experience of at least ten years in dealing with engineering geological aspects of water resources projects
5.	Hydrologist	Bachelor of Engineering or Bachelor of Technology or equivalent in Civil Engineering Preferably Master Degree in Hydrology or Hydraulic Engineering or Water Resources Development	• Experience in dealing with hydrological planning of major dam projects especially design flood estimation, review of design flood, design storm inputs, sedimentation aspects etc. of at least ten large dam projects. • Additional weightage for an expert having exposure to various relevant hydrological modelling tools, work done for dam break modelling, inundation mapping, development of emergency action plan and its implementation etc. • Professional experience of at least ten years in dealing with hydrological aspects of water resources projects
6.	Hydro- Mechanical Expert	Bachelor of Engineering or Bachelor of Technology. or equivalent in Civil Engineering or Mechanical Engineering	• Experience in rehabilitation of spillway gates, under-sluice gates, emergency gates, erection of Hydro- Mechanical works shall get additional weightage. • Demonstrate relevant experience during the last ten years prior to empanelment • Professional experience of at least ten years in designs of Hydro- Mechanical

Serial Number	Experts	Minimum qualification	Experience
(1)	(2)	(3)	(4)
			Works of large dam projects and especially works done at detail engineering projects and rehabilitation Hydro- Mechanical works done.
7.	Instrumentation Expert	Bachelor of Engineering or Bachelor of Technology. or equivalent in Civil Engineering or Instrumentation Engineering or Electronics Engineering or Master Degree in Physics	• Experience in planning and installation of all kinds of dam instruments for a minimum of three large dam projects. • Demonstrate relevant experience during the last ten years prior to applying for empanelment. Experience of reinstrumentation of old dam structures shall get additional weightage. • Professional experience of at least ten years in design of dam projects.
8.	Seismic Expert	Bachelor of Engineering or Bachelor of Technology or equivalent in Civil Engineering Preferably Master Degree in Earthquake Engineering or Structural Engineering or Physics.	• Experience in planning and designs for minimum of five large dam projects accounting for seismic loadings, having the exposure of time history series as well as seismic spectra will be an additional advantage and also work done for site specific seismic design parameters shall be given additional weightage • Demonstrate relevant experience during the last ten years prior to applying for empanelment. Experience of rehabilitation of old dam structures would get additional weightage. • Professional experience of at least ten years in design of dam projects.

ANIL JAIN, Chairman

[ADVT.-III/4/Exty./141/2025-26]

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